Job Portal Management System

Objective

The objective of this project is to design and implement a Job Portal database system that manages job postings by companies, applications submitted by candidates, and interview processes. The system focuses on maintaining structured data and handling relationships between different entities.

System Overview

The Job Portal system allows companies to post jobs and candidates to apply for those jobs. Each application is tracked with its status, and interviews are conducted for selected applications. The system ensures proper data organization using relational database concepts.

Tables

Candidate

Stores information about job seekers.
Attributes: candidate_id, name, email, qualification, experience

Company

Stores information about companies posting jobs.
Attributes: company_id, company_name, location

Job

Stores job details posted by companies.
Attributes: job_id, company_id, job_title, salary, location

Application

Stores details of job applications submitted by candidates.
Attributes: application_id, candidate_id, job_id, apply_date, status

Interview

Stores interview details for each application.
Attributes: interview_id, application_id, interview_round, interview_date, result

ER Diagram

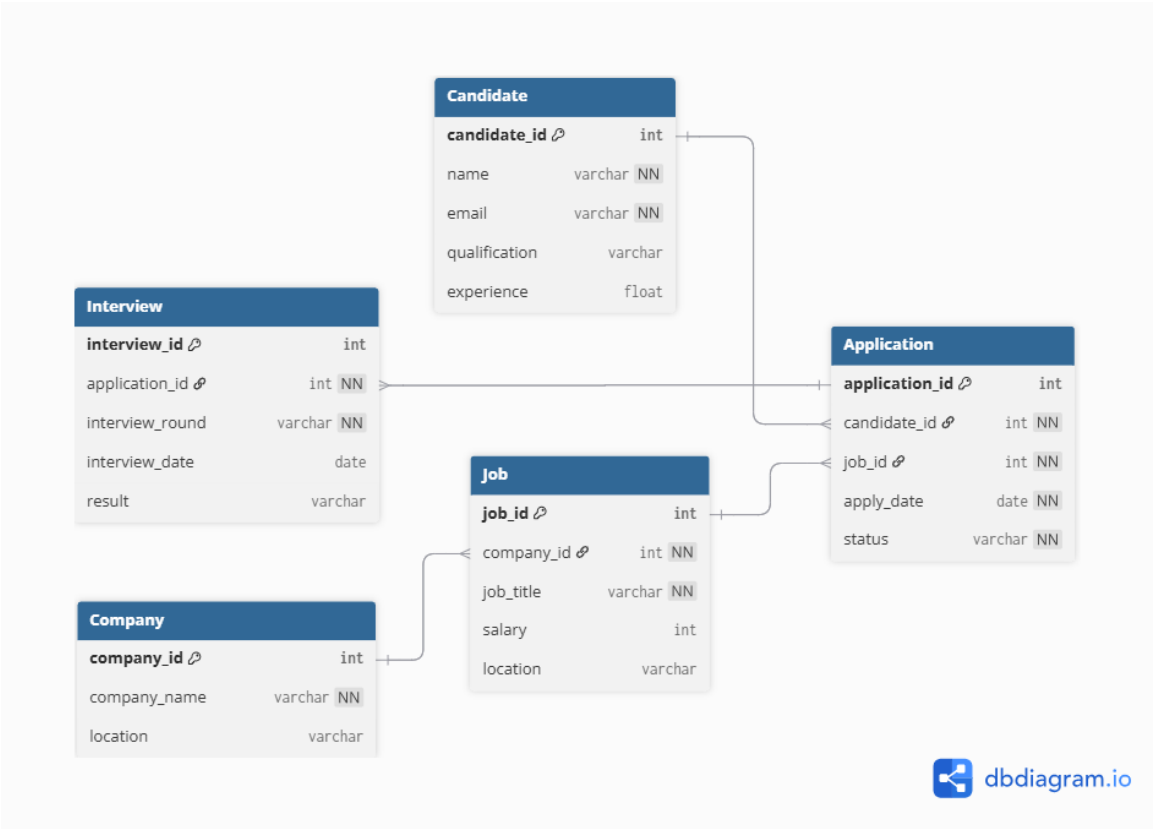

Figure: ER Diagram of Job Portal System

Relationships

1. Company → Job (1:M): One company can post multiple jobs.
2. Candidate → Application (1:M): One candidate can apply for multiple jobs.
3. Job → Application (1:M): One job can receive multiple applications.
4. Candidate ↔ Job (M:M via Application): Managed using Application table.
5. Application → Interview (1:M): One application can have multiple interviews.

Key Features

Maintains structured data using relational database design.
Handles many-to-many relationships using an intermediate table.
Tracks application status for each candidate.
Supports multiple interview rounds for each application.
Ensures data integrity using primary key and foreign key constraints.

Technologies Used

1. Oracle Database 11g
2. SQL
3. SQL*Plus
4. dbdiagram.io

Conclusion

This project shows a simple and structured database design for a job portal system. It helps manage jobs, applications, and interviews effectively.